

Background patterns — discontinuous, burst-suppression, suppressed

Three more fields the PI asked for, each labeled on its own: whether the report calls the EEG background **discontinuous**, whether it describes **burst-suppression**, and whether it calls the background **suppressed**. Same setup as the slowing work — MedGemma-27B (Q4_K_S), temperature 0, Vasily’s prompts, grammar-constrained so every answer is one of `present` / `explicitly_absent` / `not_mentioned`. Run on Fraser Health (45,545 reports) and the three Harvard hospitals — MGH, BWH, BCH (~101,600; BIDMC has no text).

Of the three, only burst-suppression has a matching column in the Harvard database (bs). Discontinuous and suppressed background have no HEEDB variable, so there we only have our own labels. For how we read Harvard’s values, see the companion note `reports/harvard_llm_labeling.md`.

How often each is called present

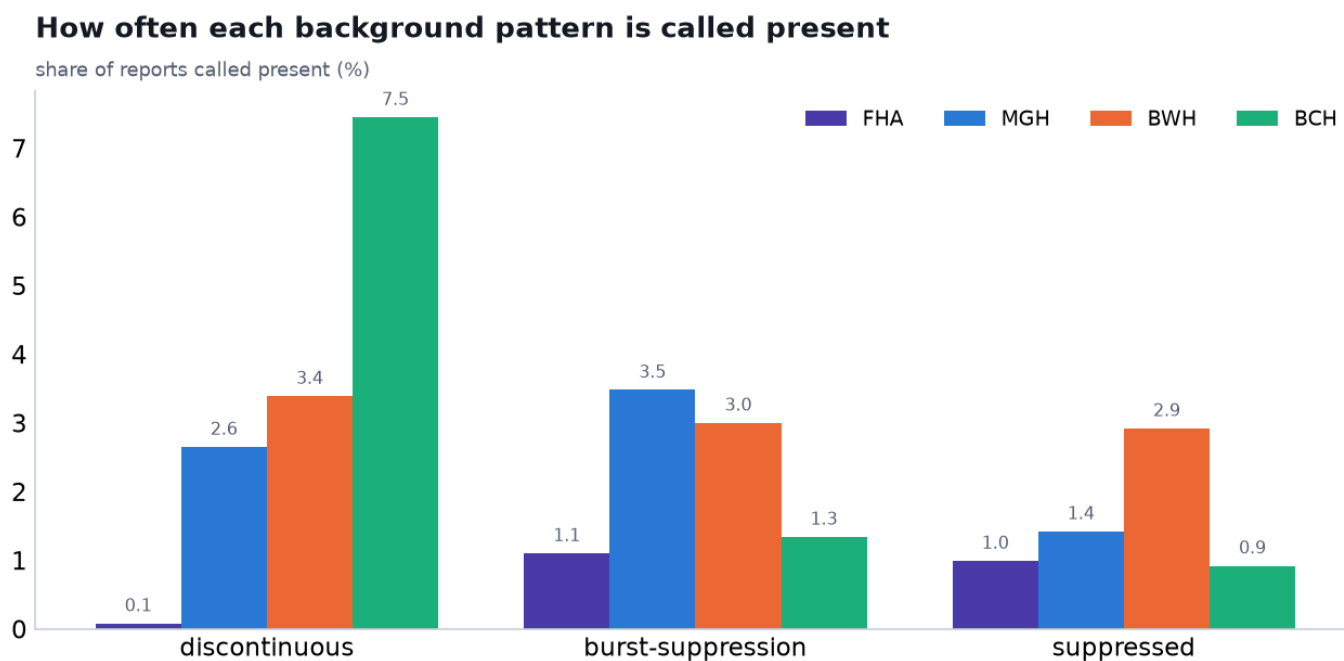


Figure 1: Present rate across datasets

All three are uncommon, as expected — these are ICU / anesthesia patterns, not routine findings.

present rate	FHA	MGH	BWH	BCH
discontinuous	0.1%	2.6%	3.4%	7.5%
burst-suppression	1.1%	3.5%	3.0%	1.3%
suppressed	1.0%	1.4%	2.9%	0.9%

The one that stands out is **discontinuous**. On Fraser Health almost no report addresses continuity at all (0.1% present, 98% not mentioned), while the Harvard reports discuss it far more often — and not just as present: they very often state the background *is* continuous. Explicit “continuous / not discontinuous” statements run to 17% at MGH, 25% at BWH and 63% at BCH, versus under 2% on Fraser Health. That’s a reporting-style difference between the cohorts (Harvard’s ICU/EMU reports routinely comment on continuity), not a disagreement about any one recording.

Burst-suppression vs Harvard's bs

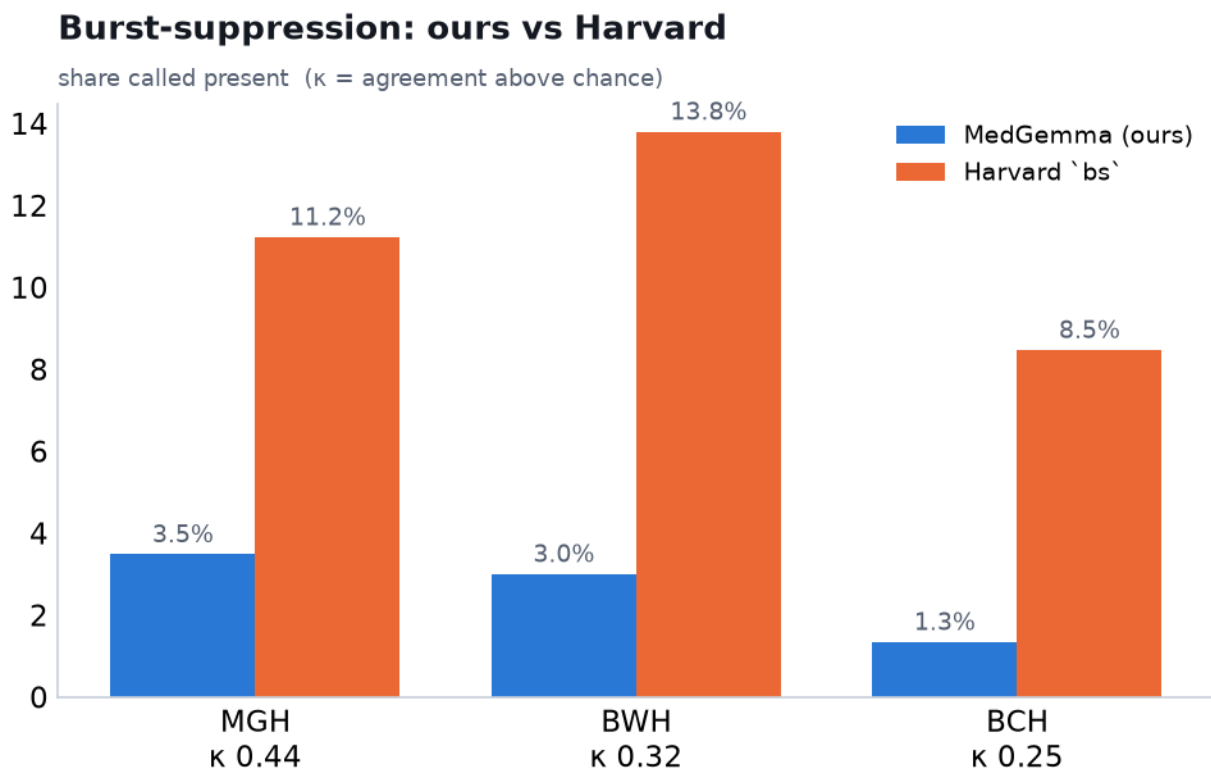


Figure 2: Burst-suppression: ours vs Harvard

	agreement	κ	κ (text only)	our recall on verified
MGH	92%	0.44	0.60	65% (231)
BWH	89%	0.32	0.50	61% (184)
BCH	93%	0.25	0.29	— (0 verified)

Two things to read here. Raw agreement looks high (89-93%), but that mostly reflects how rare the finding is — both sides say “no” on almost every report, so they agree by default; κ (agreement above chance) is the honest number, and it’s fair-to-moderate.

The bigger point is the gap in the chart: **Harvard’s bs is marked far more often than we mark it** — 11% of MGH reports, 14% at BWH, against our 1-3.5%. At MGH and BWH that is not the report narratives disagreeing with us; it’s their annotation channel. When we compare only against the labels Harvard took from the report *text* (report provenance), κ jumps to 0.60 at MGH and 0.50 at BWH. So most of the extra Harvard bs there comes from the recording’s structured annotations, which flag burst-suppression on many recordings whose written report doesn’t describe it — and which we, reading only the text, never see. BCH is the exception: its bs is mostly text-derived already, so the text-only κ barely moves (0.25 $\rightarrow$ 0.29) and the gap there (8.5% vs our 1.3%) is a real under-call on the reports themselves.

On the reports an expert had verified as burst-suppression, we catch about 61-65%. Part of the miss is the same channel issue (some verified labels are annotation-derived and not in the text), so this is a floor rather than a clean recall.

Discontinuous and suppressed background

There’s no Harvard column for either, so these are our labels only, no comparison:

	present	explicitly_absent	not_mentioned
discontinuous — FHA	0.1%	1.8%	98.1%
discontinuous — Harvard (MGH/BWH/BCH)	2.6-7.5%	17-63%	30-80%
suppressed — FHA	1.0%	1.7%	97.4%
suppressed — Harvard	0.9-2.9%	0.1-0.7%	97-99%

Suppressed background behaves like a normal rare finding across both datasets. Discontinuous, as noted, is the one the Harvard reports actively comment on far more than Fraser Health’s do.

Caveats

- About 1.3% of Harvard reports are too long for the context window and returned no label; they’re left out of the counts.
- Only burst-suppression can be compared against Harvard; discontinuous and suppressed have no HEEDB variable (nearest is low voltage, a different thing).
- The verified check is positives-only (no verified “absent”) and has no BCH data.

Reproduce

```
# labeling (on the cluster)
bash slurm/submit_background.sh
bash slurm/submit_background_harvard.sh
# distributions, comparison, charts
python -m analysis.background_dist
python -m analysis.background_vs_harvard
python -m analysis.background_chart
```

```
# Fraser Health, all three fields
# MGH / BWH / BCH, all three fields

# results/background/dist_summary.json
# results/harvard/burst_vs_harvard.json
# the two figures
```